Reproducing Spectral Shrunk Clustering: Faithful Stationary Points and Unreproduced Gains

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Abstract

This course report reimplements Spectral Shrunk Clustering (SSC) of Chang, Nie, Ma, Yang, and Zhou from the printed equations, without author code. After a typesetting correction, the faithful linear system is the stationary point of Equation (8), $A = S + 2\gamma\widetilde{L}$ (variant E8). Printed Equation (13) omits that factor of two (variant E13). Spectral clustering (SC) and SSC now share one configurable Equation (4) embedding F. No named dataset is fully protocol-reproduced. AR was not downloaded; MSRA50 was not found; a Binary Alphadigits 26×30 draw is UNVERIFIED; UMIST archives yield 565 unique cropped images, not 564. On AT&T ORL (`att92`), faithful E8 accuracy is 57.3 ± 3.8 ($n = 50$) versus the published 76.0 ± 1.6 (-18.7 points). The gap is already present when clustering F (56.2 ± 4.2 versus published SC 72.7). Shrinkage does not help. A k-means++ score of 78.3 ± 1.3 on the same saved F is an ambiguity (B), not a reproduction. SPLS remains unimplemented. No new theorem is claimed.

1 Introduction

Spectral clustering [Shi and Malik, 2000, von Luxburg, 2007] builds a similarity graph, takes Laplacian eigenvectors, and runs K-means in that embedding. COMP3314 contrasts that neighbourhood geometry with PCA, which keeps high-variance axes, and with LDA, which uses class scatter. Chang et al. argue that a graph built on the original points x_i need not be the right regularizer for the embedded points f_i. Spectral Shrunk Clustering (SSC) therefore computes a normalized-cut embedding F, rebuilds a k-nearest-neighbour (kNN) graph on F, and solves a convex shrinkage problem for G before the final K-means [Chang et al., 2015].

This report keeps four tasks separate. First, we reimplement SSC from the equations and record every unspecified choice as an assumption. Second, we obtain datasets only from files with a URL, date, and SHA-256; missing names stay missing. Third, published ACC/NMI are labelled "Chang et al. (2015)" and are never copied into an Our Reproduction cell. Fourth, we replace the shrinkage graph by mutual kNN as a Category C extension, not as a new method, and report a negative difference. This is an HKU COMP3314 project, not a NeurIPS submission, and it states no novel theorem.

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2 Background and SSC

Lloyd K-means is the last discretization in SSC and in the spectral baseline [Lloyd, 1982]. PCA+K-means keeps variance; unsupervised LDA+K-means of Ding and Li [2007] uses current cluster labels, never ground truth, and has rank at most $c - 1$. Spectral clustering uses the symmetric normalized Laplacian of Equation (4) in Chang et al.,

$$L_n = I - D^{-1/2} A D^{-1/2}. \tag{1}$$

The paper sets $k = 5$ for every spectral method. We take δ^2 as the median of positive directed kNN squared distances (unspecified in the paper; Category B) and undirected edges as the union of the two directed relations [von Luxburg, 2007]. F holds the c eigenvectors for the smallest eigenvalues. Isolated vertices get $D_{ii}^{-1/2} = 0$. Eigenvectors are *not* row-normalized; Ng–Jordan–Weiss row-norm [Ng et al., 2001] is Category C even when ACC is higher. Local learning clustering [Wu and Schölkopf, 2006] and adaptive neighbours [Nie et al., 2014] are different methods. Pattern shrinking [Hou et al., 2013] is cited as SPLS (the source acronym is SLPS); we did not obtain Section 3 of that paper and keep SPLS as `NOT_IMPLEMENTED`. LDA-SC is named but not defined; it stays `APPROXIMATE` and is omitted from the faithful table.

Shrinkage. A second Gaussian kNN graph W is built on the rows of F. The printed convex problem is

$$\min_G \|G - F\|_{2,1} + \gamma \sum_{ij} W_{ij} \|g_i - g_j\|_2, \tag{2}$$

with $\|H\|_{2,1} = \sum_i \|h_i\|_2$ over rows. The sum is over ordered pairs. Equation (10) writes an embedding-dimension diagonal for S, which cannot sit in the $n \times n$ trace of Equation (9) or add to the $n \times n$ Laplacian. We use the sample-wise $S_{ii} = 1/(2\max(\|h_i\|_2, \varepsilon))$. Algorithm 1 line 10 replaces $\widetilde{L}$ by $\widetilde{W}$ and F by X; we do not execute that line.

Equation (8) versus Equation (13). For symmetric W, $\sum_{ij} \widetilde{W}_{ij} \|g_i - g_j\|_2^2 = 2\,\mathrm{Tr}(G^\top \widetilde{L} G)$. Differentiating (2) yields the stationary system $(S + 2\gamma\widetilde{L})G = SF$. Printed Equation (13) uses $\gamma\widetilde{L}$ instead of $2\gamma\widetilde{L}$, so it is the stationary point of (2) with graph coefficient $\gamma/2$. We treat that factor of two as a typesetting correction, not a contribution. Variant **E8** ($A = S + 2\gamma\widetilde{L}$) is the faithful reconstruction of printed Equation (8). Variant **E13** ($A = S + \gamma\widetilde{L}$) is the printed update and is Category B relative to Equation (8). A six-point CVXPY/CLARABEL check is in Appendix A. Default $G_0 = F$. We stop on a relative Frobenius step of 10^{-6} or after 300 iterations. Labels come from K-means on G and are never written back into W or F.

3 Reproduction setup

Results are tagged A (paper-faithful after the corrections above), B (unspecified reconstruction), or C (non-faithful diagnostic). A closer number does not promote B or C to A.

SC and SSC now share one `SpectralGraphConfig`: $k = 5$, union kNN, median directed bandwidth, Equation (4), no row-norm. After a largest-magnitude-positive sign convention, the two F matrices match elementwise (maximum absolute difference 0) on Olivetti and on `att92`. There is no $n > 1500$ backend switch.

Default `load_dataset("ORL")` remains scikit-learn Olivetti 64×64 bilinearly resized to 32×32 [Samaria and Harter, 1994, Pedregosa et al., 2011]. Round 2 original-source ORL is version `att92`: Cambridge AT&T 400 PGM files at 92×112, bilinear 32×32. JAFFE is the verified Zenodo TIFF set, 213 images, resized to 26×26 [Lyons et al., 1998, 2025]; redistribution is forbidden. Default YaleB is Deng Cai's `YaleB_32x32.mat` (2414×38); version `cropped2414` is a Wayback CroppedYale zip with 2414 non-ambient $(192, 168)$ images, not claimed pixel-identical to the mat [Georghiades et al., 2001, Lee et al., 2005]. USPS is the LIBSVM 9298×256 pair [Hull, 1994]. A Zenodo AR archive was inspected and *not* downloaded (unofficial; claimed CC-BY conflicts with the official no-redistribution term). Binary Alphadigits was obtained; a 26×30 letter subset with seed 0 is an UNVERIFIED reconstruction, not the paper BinAlpha. MSRA50 was not found. UMIST

Table 1: Hungarian ACC and geometric NMI (%). Paper cells are Chang et al. (2015), reported only. Our SC and E8 are protocol A. k-means++ and row-norm are not in this table. Run records: `results/round2/`.

Set (n)	Metric	Paper SC	Our SC (A)	Paper SSC	Our E8 (A)
ORL att92 (50)	ACC	72.7±2.3	56.2±4.2	76.0±1.6	57.3±3.8
	NMI	85.8±1.9	80.5±2.3	88.6±1.5	81.1±2.2
JAFFE (50)	ACC	76.1±1.6	68.5±11.2	81.2±1.6	68.4±11.1
	NMI	80.2±0.9	83.0±5.7	84.3±1.6	83.0±5.7
YaleB (10)	ACC	34.8±1.4	20.1±1.2	49.8±1.6	19.5±1.5
	NMI	55.6±1.6	29.9±0.9	46.8±1.3	29.3±1.2
USPS (3)	ACC	64.3±1.4	63.8±1.7	75.5±1.9	60.7±4.0
	NMI	69.5±1.6	77.4±3.9	79.8±1.6	76.8±3.1

Mean±sample std over completed random K-means trials; n is the actual count, never padded to 50. Paper cells are *not* Our Reproduction. Protocol A: Equation (4), k=5, union kNN, median directed bandwidth, raw eigenvectors, γ=1, G_0=F, random K-means, n_{init}=1. ORL is AT&T 92×112 bilinear 32×32 (`att92`). JAFFE is verified Zenodo. YaleB is Deng Cai 32×32 (n=10). USPS is LIBSVM (n=3). E8 is $A = S + 2\gamma\widetilde{L}$. E13, k-means++, and row-norm are omitted here: they are B or C, not faithful SSC. AR, UMIST (no 564-image file), BinAlpha (UNVERIFIED reconstruction only), MSRA50, and SPLS are N/A — not reproduced.

cropped files have 575 images and 565 unique pixel arrays, not 564 [Graham and Allinson, 1998]. No dataset is `FULL_PROTOCOL_REPRODUCED`. No substitute collections were renamed.

Protocol: $k = 5$, $\gamma = 1$ frozen without labels, union kNN, median bandwidth, raw eigenvectors, $G_0 = F$, IRLS budget 300. F is solved once; K-means uses distinct seeds, `init="random"`, $n_{\text{init}} = 1$. We report mean±sample standard deviation over completed trials and never pad to 50. ACC is Hungarian matching [Munkres, 1957]; NMI is the Strehl–Ghosh geometric mean [Strehl and Ghosh, 2002]. Hardware: 8 CPU cores, 16 GB RAM, Python 3.13.3, no GPU. Code: `https://github.com/SelinaMai-SYM/COMP-3314-G23.git` branch `v1`. Records: `results/round2/`.

4 Results and analysis

Table 1 is the only main-text comparison. Paper columns quote Chang et al.; they are not Our Reproduction. Trial counts are actual: ORL/JAFFE $n = 50$, YaleB $n = 10$, USPS $n = 3$.

The ORL gap appears before IRLS. Historical Olivetti SC was 54.2 ± 3.1 ($n = 50$; Track A/B `...a86c9d70`, `...263aa38a`). After unifying F, the same Olivetti protocol is 53.9 ± 3.0 (`D01_OLIVETTI_SC`). Original-source `att92` SC is 56.2 ± 4.2 versus published SC 72.7 (-16.5 points). Faithful E8 on that F is 57.3 ± 3.8 versus published SSC 76.0 (-18.7 points). E13 is 56.7 ± 4.1. Shrinkage moves ACC by about one point. SC does not run IRLS, so a solver bug cannot explain the SC gap.

Shrinkage did not help. On the same embeddings, K-means(F) versus E8 K-means(G) is 56.2 vs 57.3 (ORL `att92`, $n = 50$), 68.5 vs 68.4 (JAFFE, $n = 50$), 20.1 vs 19.5 (YaleB Deng Cai, $n = 10$), 18.1 vs 17.4 (CroppedYale, $n = 10$), and 63.8 vs 60.7 (USPS, $n = 3$). E8 attains a lower Equation (8) value than E13 on ORL, but clustering does not follow.

Faithful versus numerically closer. The most faithful reconstruction is E8 (A). k-means++ on the saved `att92` F is 78.3 ± 1.3 (B; best inertia ACC 79.0). That number exceeds published SC 72.7 and SSC 76.0 and is *not* a reproduction: the paper writes random initialization and mean±std. Random K-means' best inertia on the same F has ACC 62.3, not 72.7. Row-norm is 67.7 ± 3.1 ($n = 5$, C). Neither B nor C enters Table 1.

5 Extension and discussion

The extension replaces union kNN W on F by mutual (AND) kNN with the same Gaussian weights frozen before IRLS [Brito et al., 1997, Maier et al., 2009]. Convexity of (2) is unchanged; no new theorem is claimed. On JAFFE at $\gamma = 1$, $k = 5$, $n = 50$, mutual E8 ACC is 68.4 ± 11.2 versus union E8 68.4 ± 11.1 (410 vs 655 edges; 20 vs 9 components). The outcome is negative, not a win, and it does not close the ORL gap.

The modelling claim—regularize the embedding with a graph on the embedding—is independent of the typesetting errors. The published optimization story is weaker: Equation (10), Algorithm 1 line 10, and the Eq. (8)/Eq. (13) coefficient all change the linear system. Label-informed "best results" on a γ grid is not a fair unsupervised tune; this round froze $\gamma = 1$. Four dataset names remain unmatched at the paper counts. SPLS, described as the second-best baseline, has no Our Reproduction numbers. K-means++ (B) moves ACC, which shows that the unspecified discretization matters, not that 2015 used k-means++.

Limitations in force: no author code; median bandwidth and $G_0 = F$ are assumptions; YaleB $n = 10$ and USPS $n = 3$; IRLS often unconverged on ORL/YaleB at 300 steps (1000 steps on a 5-seed ORL pilot converged without raising ACC); Olivetti and Deng Cai defaults are not claimed sample-identical to 2015; face images collected by others are gitignored and not redistributed. Author-contact questions remain listed in the repository and were not emailed.

6 Conclusion

We reimplemented SSC with a shared Equation (4) F, an explicit E8/E13 split, and extra ORL/YaleB archives that do not change default loaders. Shrinkage does not help on the obtained files. Faithful E8 does not reproduce the ORL 76% accuracy; the disagreement is already present in random K-means on F. The most faithful reconstruction and a k-means++ number that happens to sit near the published table are not the same object. The mutual-kNN extension is negative. SPLS is not implemented. The honest product is the protocol, the corrections, the missing data, and those negative findings.

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HKU COMP3314 course report. No external funding is claimed. Names and emails require confirmation. Code is intended for `https://github.com/SelinaMai-SYM/COMP-3314-G23.git` on branch `v1`.

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A Equation audit

This appendix records the locked E8/E13 split. It is a correction of Chang et al., not a claimed theorem.

Laplacian identity. For symmetric nonnegative W and combinatorial $\widetilde{L} = \widetilde{D} - \widetilde{W}$,

$$\sum_{ij} \widetilde{W}_{ij} \|g_i - g_j\|_2^2 = 2\,\mathrm{Tr}(G^\top \widetilde{L} G).$$

A two-point check with $G = (0.2, 0.8)^\top$, $W_{01} = W_{10} = 1$, and distance 0.6 gives $\sum_{ij} W_{ij}\|g_i - g_j\|_2^2 = 0.72$, $\sum_{ij} \widetilde{W}_{ij}\|g_i - g_j\|_2^2 = 0.6$, and $\mathrm{Tr}(G^\top \widetilde{L} G) = 0.3$.

Gradient of Equation (2)**.** For $h_k \neq 0$ and unused self-loops,

$$\frac{\partial}{\partial g_k}\|h_k\|_2 = 2S_{kk}(g_k - f_k), \qquad \frac{\partial}{\partial g_k}\sum_{ij} W_{ij}\|g_i - g_j\|_2 = 4(\widetilde{L}G)_k.$$

Setting the gradient to zero and dividing by 2 yields $(S + 2\gamma\widetilde{L})G = SF$. Printed Equation (13) omits the 2. The same factor appears in the IRLS majorizer of Equation (15) in the paper, because that quadratic uses $\sum_{ij} \widetilde{W}_{ij}\|g_i - g_j\|_2^2 = 2\,\mathrm{Tr}(G^\top \widetilde{L}G)$.

Independent convex check. On a synthetic chain with $n = 6$, $c = 2$, $\gamma = 0.8$, seed 1, CVXPY 1.9.3 / CLARABEL 0.11.1 reports status `optimal`. E8 matches the minimizer of printed Equation (2) (objective 4.021064576828 vs 4.021064576768; largest absolute entry difference 1.07×10^{-6}). E13 matches the $\gamma/2$ instance (objective 3.843869772479 vs 3.843869772520) and does *not* minimize printed Equation (2) (value 4.536864561036 at that G). A two-block graph repeats the split. These are solver-audit values, not clustering scores, and they are not copied from Chang et al.

Four unequal scalars. The unsmoothed objective, the ε-smoothed objective, the IRLS quadratic surrogate, and the recomputed fixed-point residual are four different numbers. A small IRLS step can coexist with a large residual.

B Dataset files

Checksums were recorded on 2026-10-21 and live in the named manifests. Raw images are gitignored. Default loaders are unchanged. No dataset is `FULL_PROTOCOL_REPRODUCED`.

ORL. Default: figshare Olivetti (`benchmarks/orl/manifest.yaml`). Extra `att92`: Cambridge AT&T 400 images at 92×112 (`manifest_att92.yaml`).

JAFFE. Zenodo record 14974867 (`benchmarks/jaffe/manifest.yaml`). Non-commercial research only; no redistribution.

YaleB. Default: Deng Cai `YaleB_32x32.mat` (`manifest.yaml`). Extra `cropped2414`: Wayback CroppedYale zip, 2414 non-ambient $(192, 168)$ PGM (`manifest_cropped.yaml`). Not claimed pixel-identical to the mat. Live UCSD zips returned HTTP 404.

USPS. LIBSVM `usps.bz2` / `usps.t.bz2` (`benchmarks/usps/manifest.yaml`).

Unavailable at paper counts. AR: official hosts failed; Zenodo `ARDB_Full.zip` (HEAD 200, 120587997 bytes) was not downloaded because the uploader is unofficial and the claimed CC-BY conflicts with the official no-redistribution term. UMIST archives on disk are 1012 uncropped or 575 cropped images; dropping 10 pixel-duplicate cropped files yields 565, not 564. BinAlpha: Binary Alphadigits parent SHA-256 `3ee9392e64e4e3bb7b35502b69bf2555a0b4197c1b31048c4189b84bc0bc6ea7`; the 26×30 seed-0 draw is UNVERIFIED and is not called the paper set. MSRA50: literature pointer to He et al. without a file; MSRA-B/10K/TD500 were not renamed.